



UNIVERSITY^{AT}ALBANY
STATE UNIVERSITY OF NEW YORK

CSI 436/536 (Spring 2026)

Machine Learning

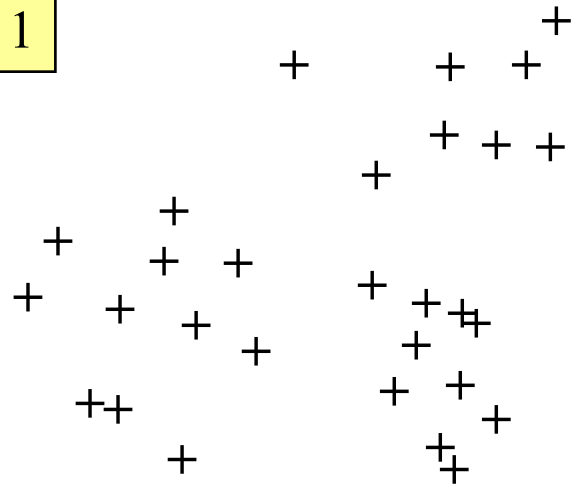
Lecture 18: Dimension Reduction

Chong Liu

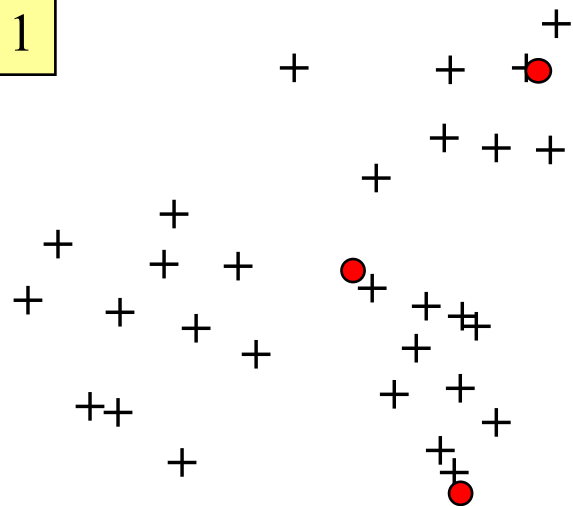
Department of Computer Science

Apr 15, 2026

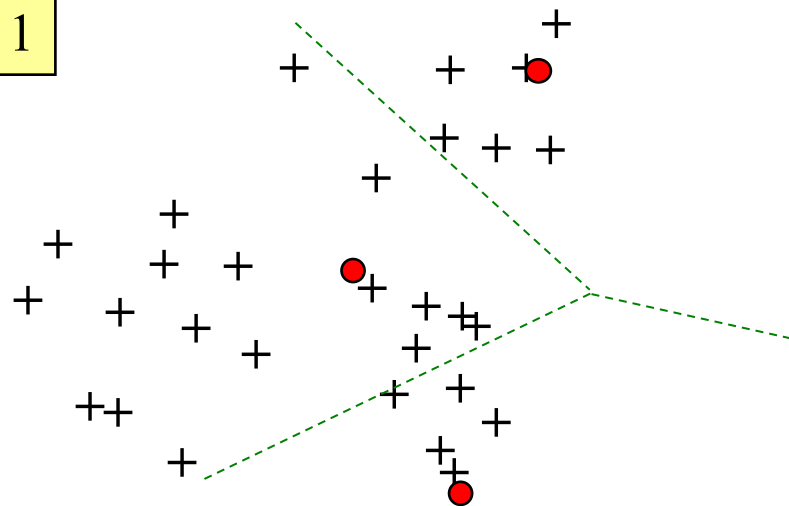
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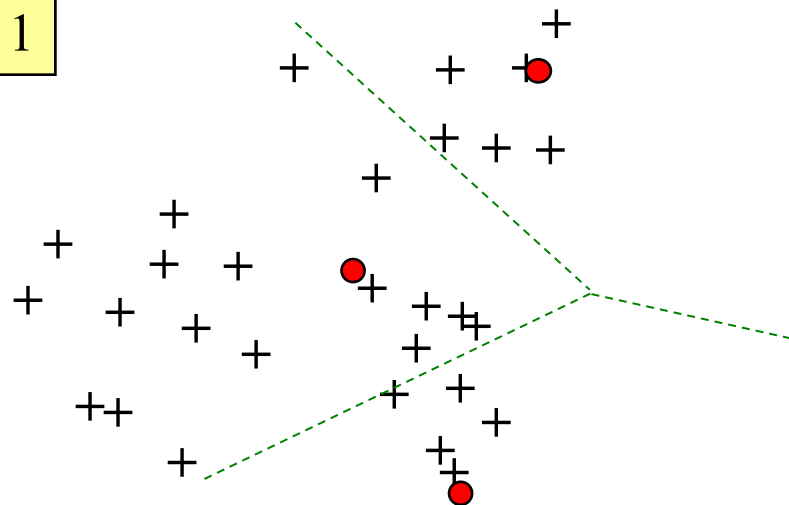
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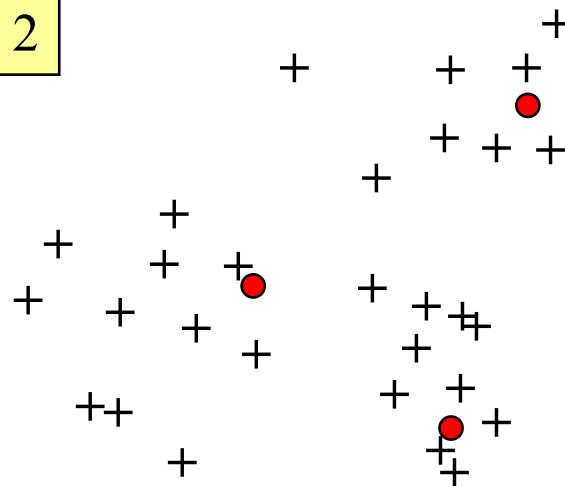
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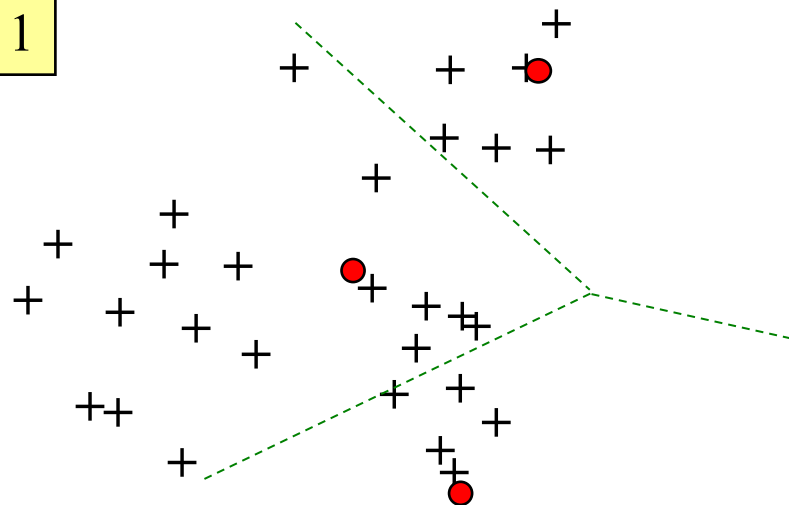
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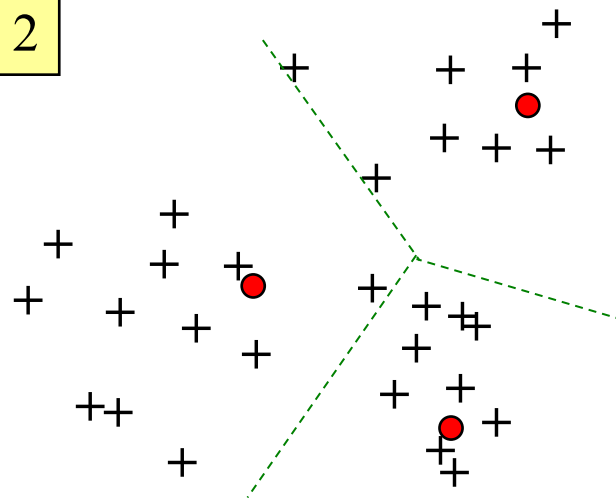
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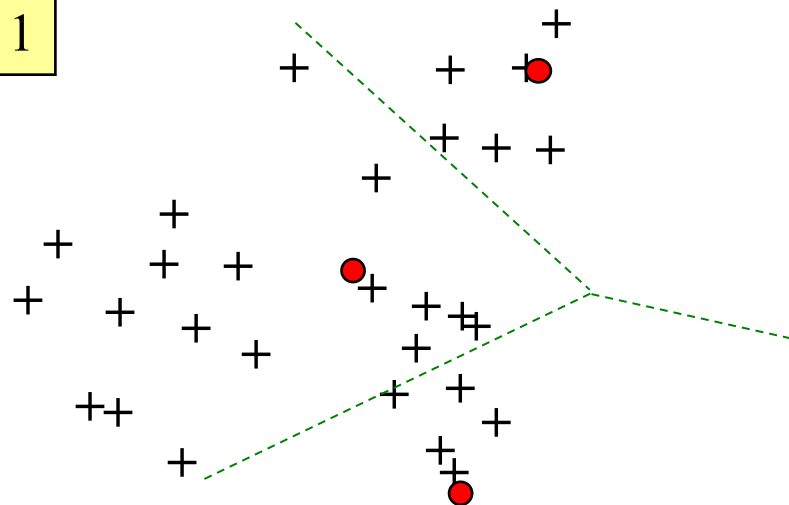
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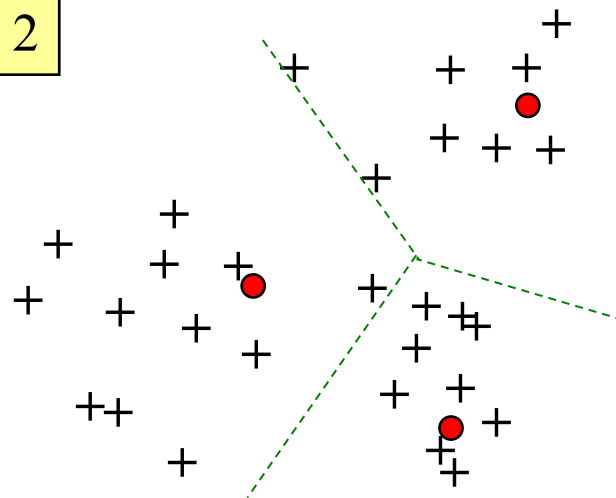
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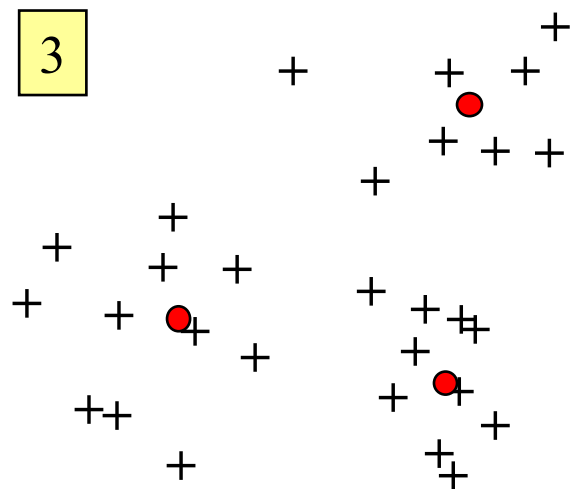
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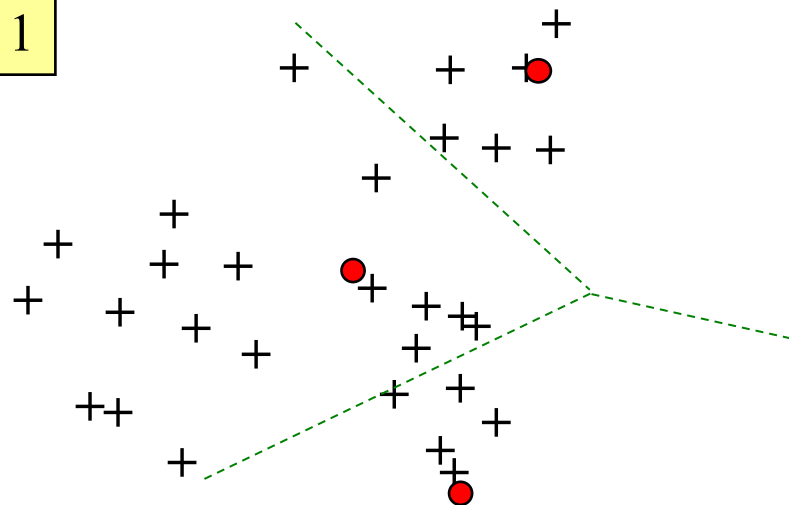
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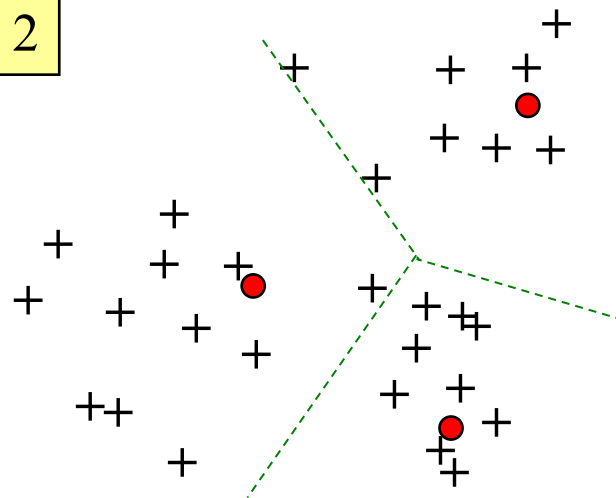
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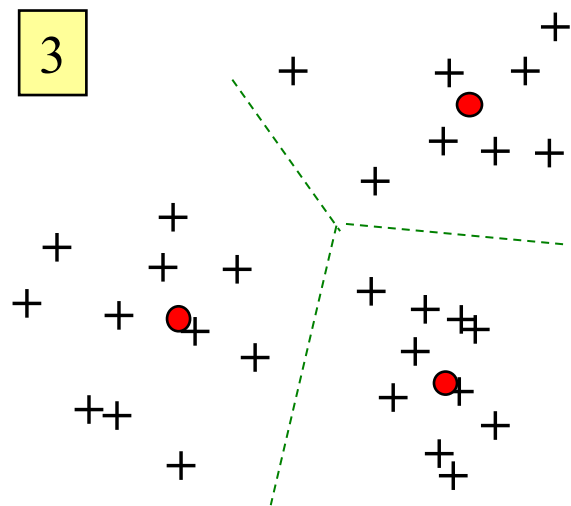
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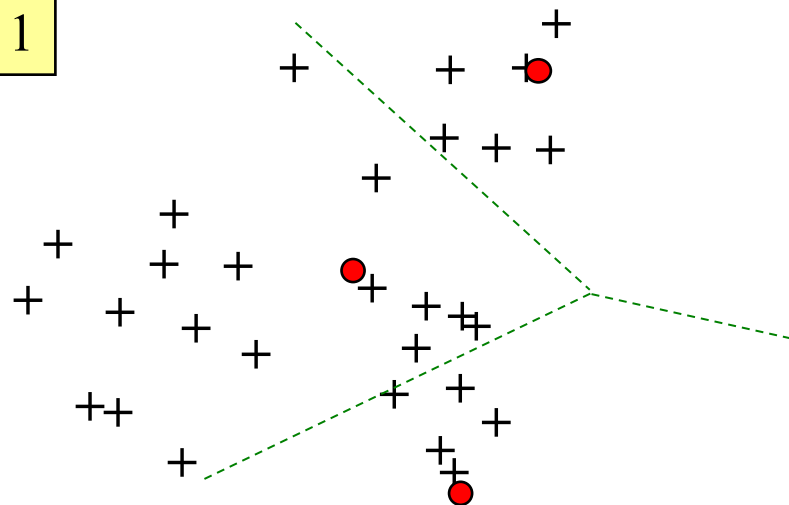
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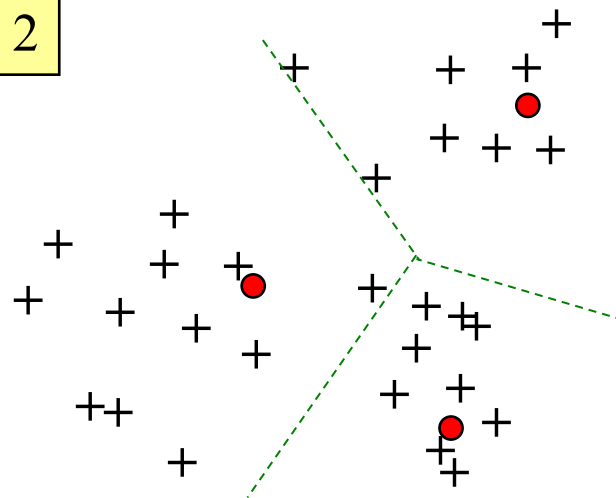
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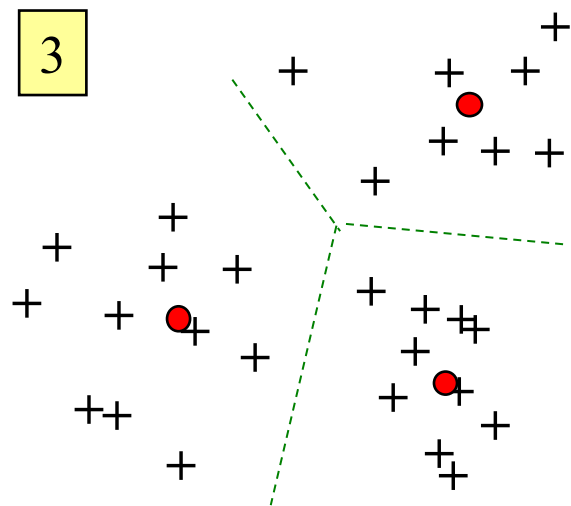
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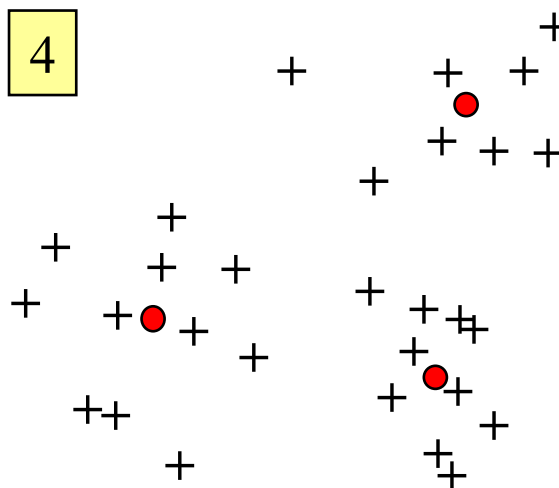
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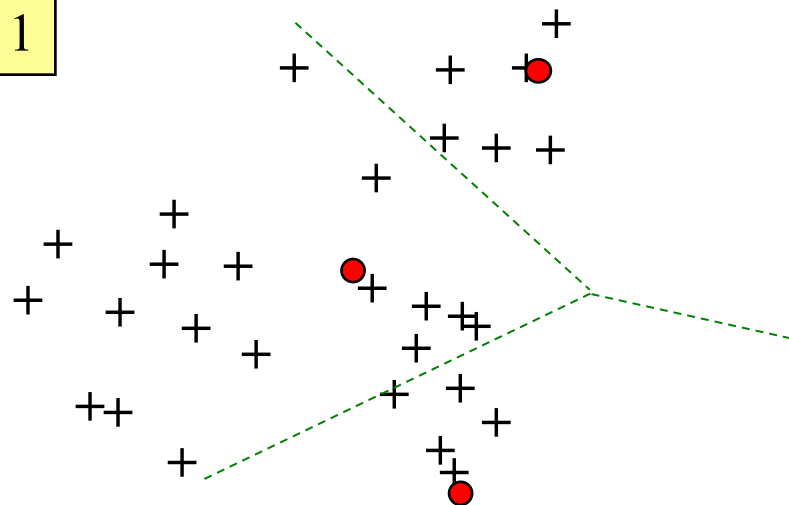
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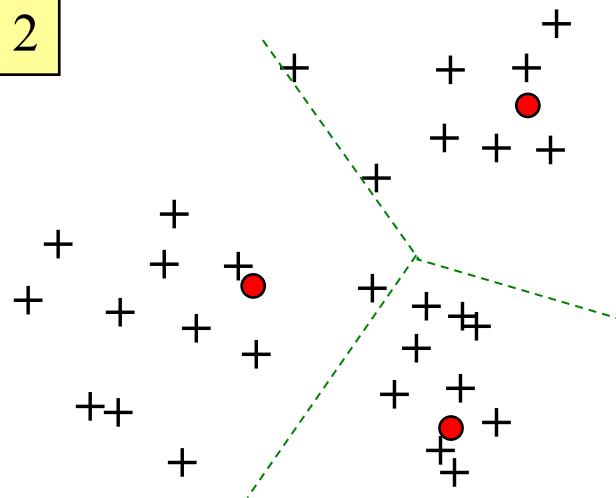
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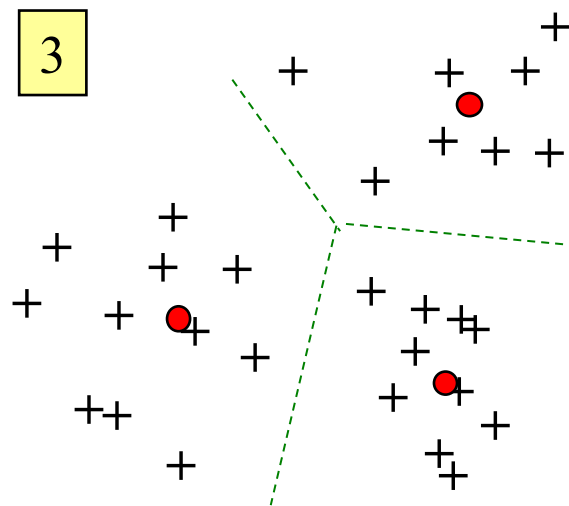
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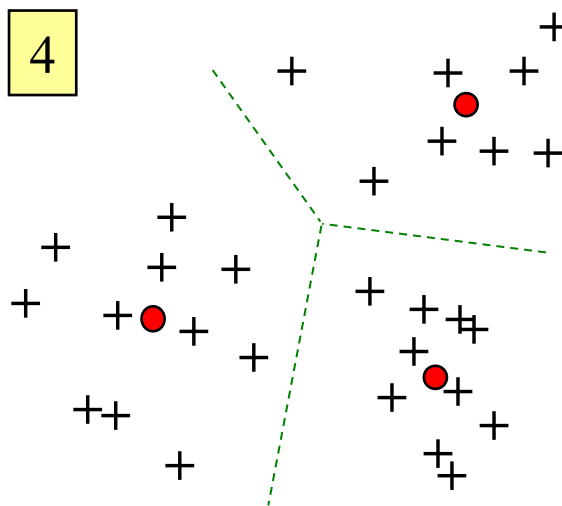
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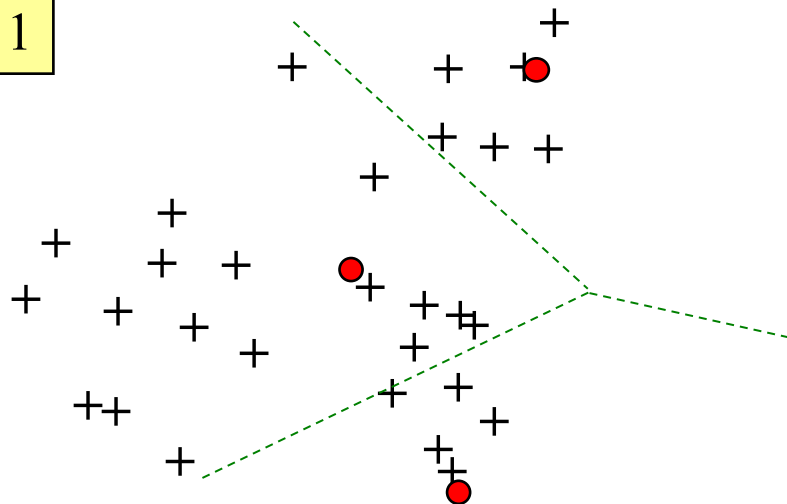
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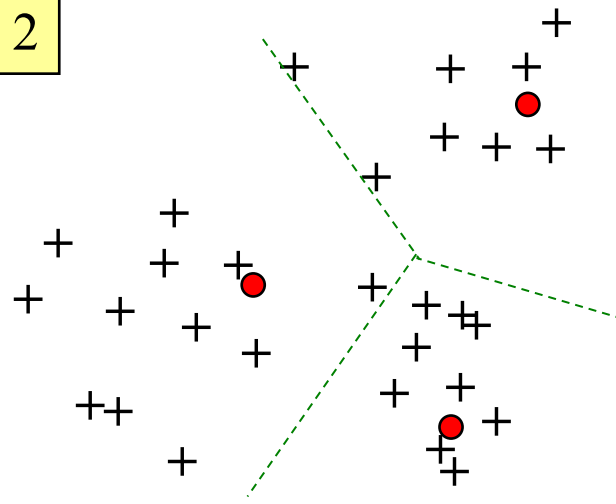
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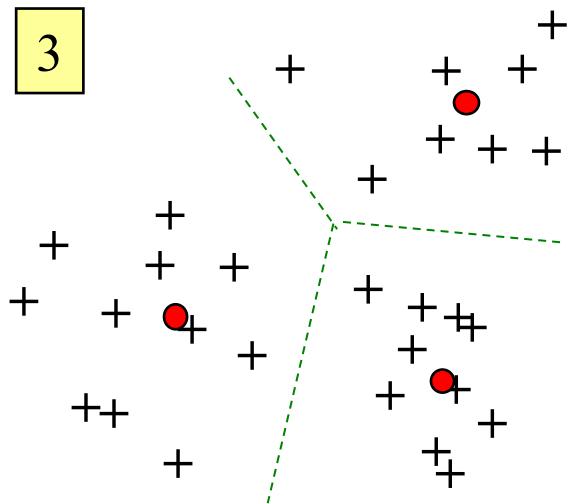
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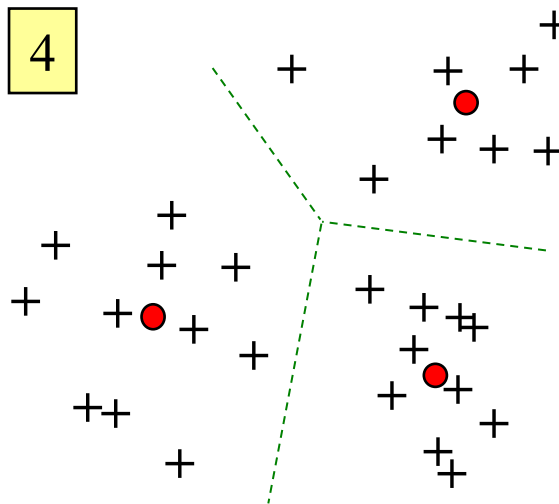
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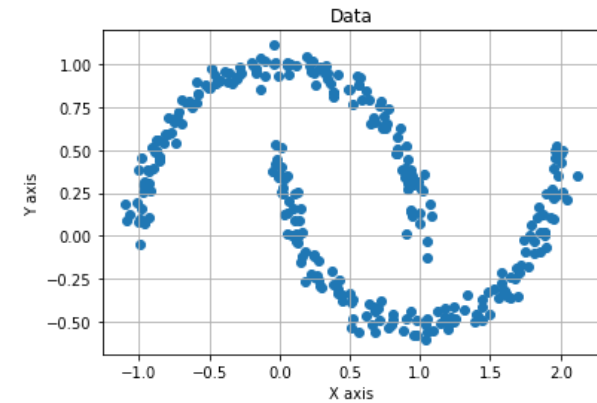
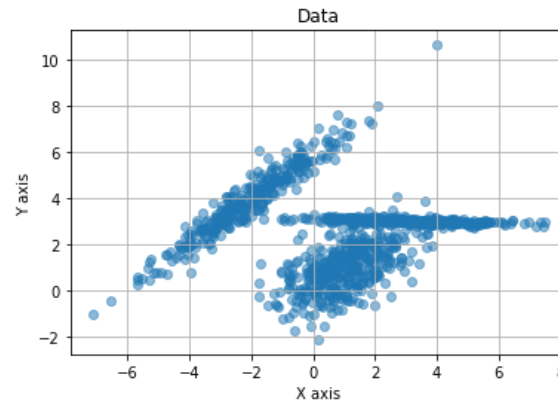
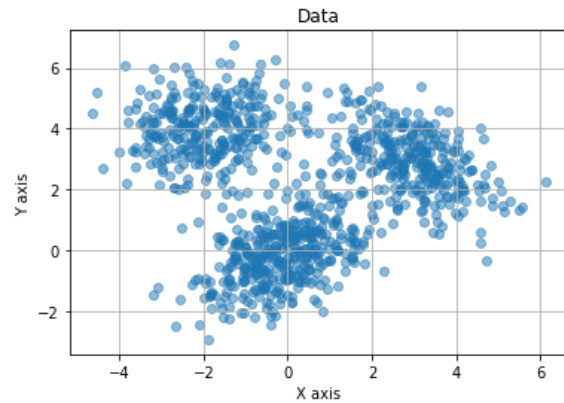
No change – finished!

Recap: Clustering

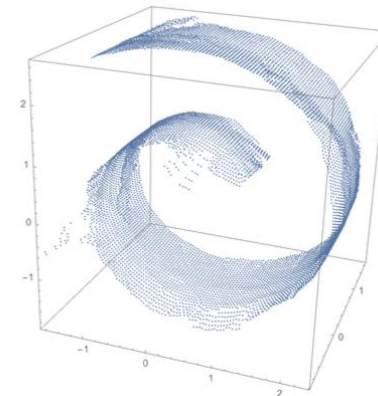
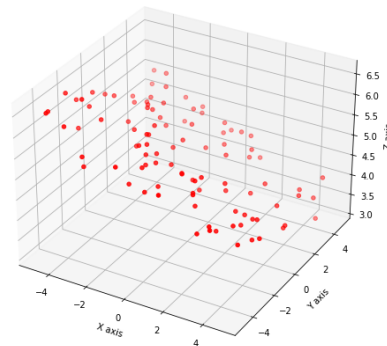
- K-means algorithm
 - Assign hard labels to data points
 - How does it work?
 - Alternating makes updates
 - Which distance function to use?
 - How many cluster centers (centroids) to choose?
 - How to initialize the centroids?
- Gaussian mixture models
 - Assign soft labels to data points
 - A probabilistic model for clustering

Recap: Two broad categories of unsupervised learning (1) Clustering **(2) Dimension reduction**

- Clustering aims at finding a partition of the data that makes sense.



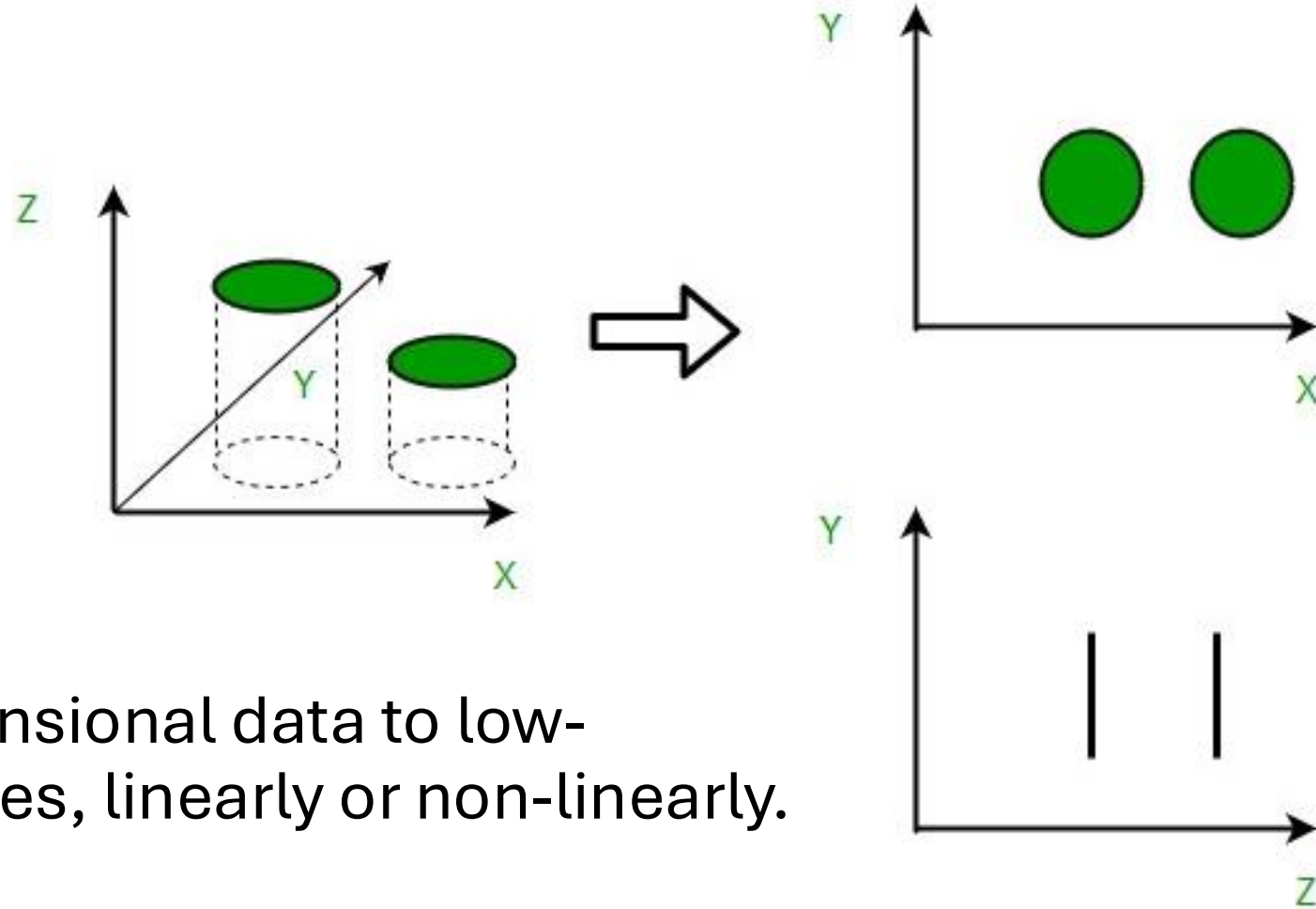
- Dimension reduction aims at identifying a more compact representation of data



Today

- Why dimension reduction?
- Linear dimension reduction
 - Principal Component Analysis (PCA) algorithm
- Non-linear dimension reduction

What is dimension reduction?



- Project high-dimensional data to low-dimensional spaces, linearly or non-linearly.

Why dimension reduction

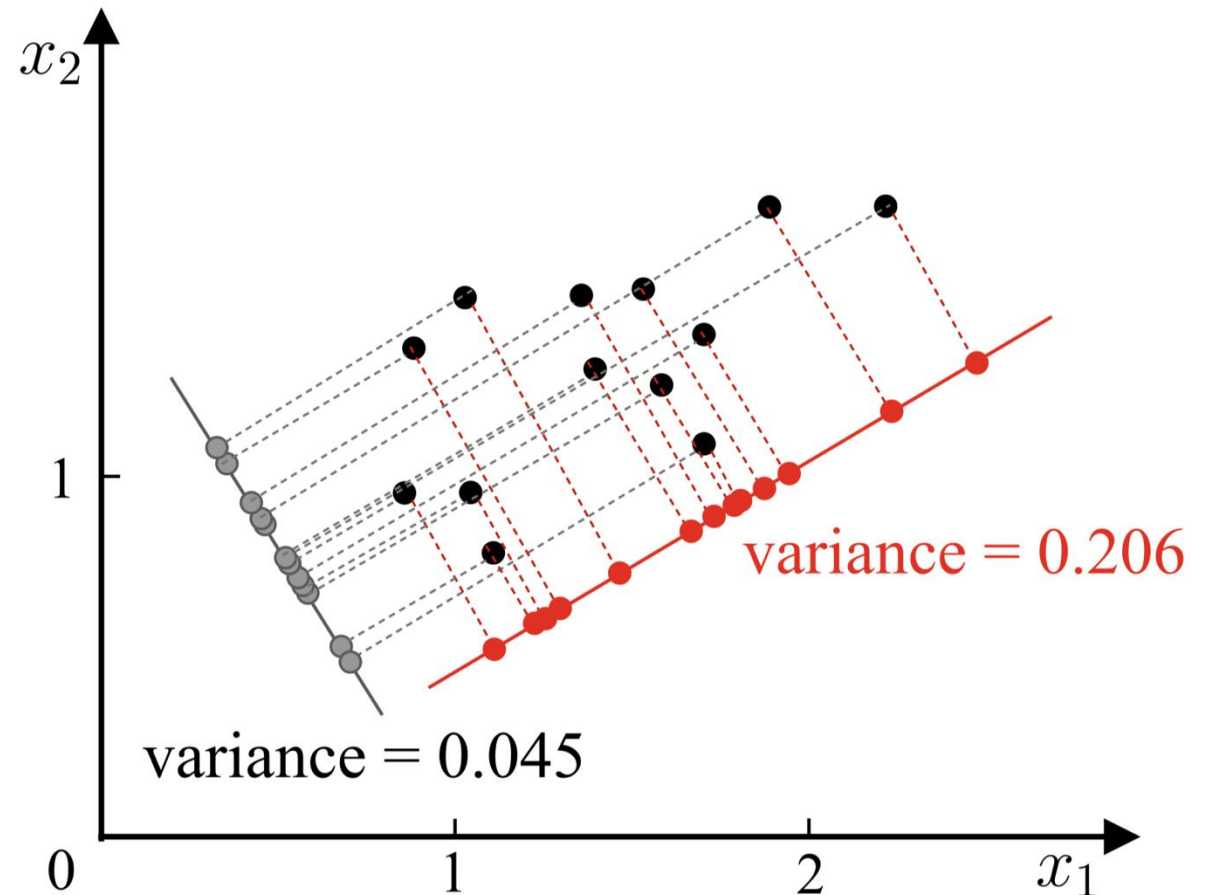
- Computation and memory efficiency
 - Reduce file size
- Statistical efficiency (fewer features to learn):
 - “Curse of dimensionality”
- Fewer features are easier to understand. It can help identifying hidden causes factors.
- Often data are high-dimensional but the physics mandate that they should be lying on a low-dimensional subspace.

Linear dimension reduction

- Input: $X \in R^{d \times m}$
 - Number of data points: m
 - Number of features: d
- Output: $X' = WX \in R^{d' \times m}$
 - Projection matrix $W \in R^{d' \times d}$
 - Discussion: Does a random W works as a valid projection matrix?
- What is a good X' ?
 - As long as it **makes sense** to your task
 - A good low-dimensional representation that is **good for further process** (e.g., classification)

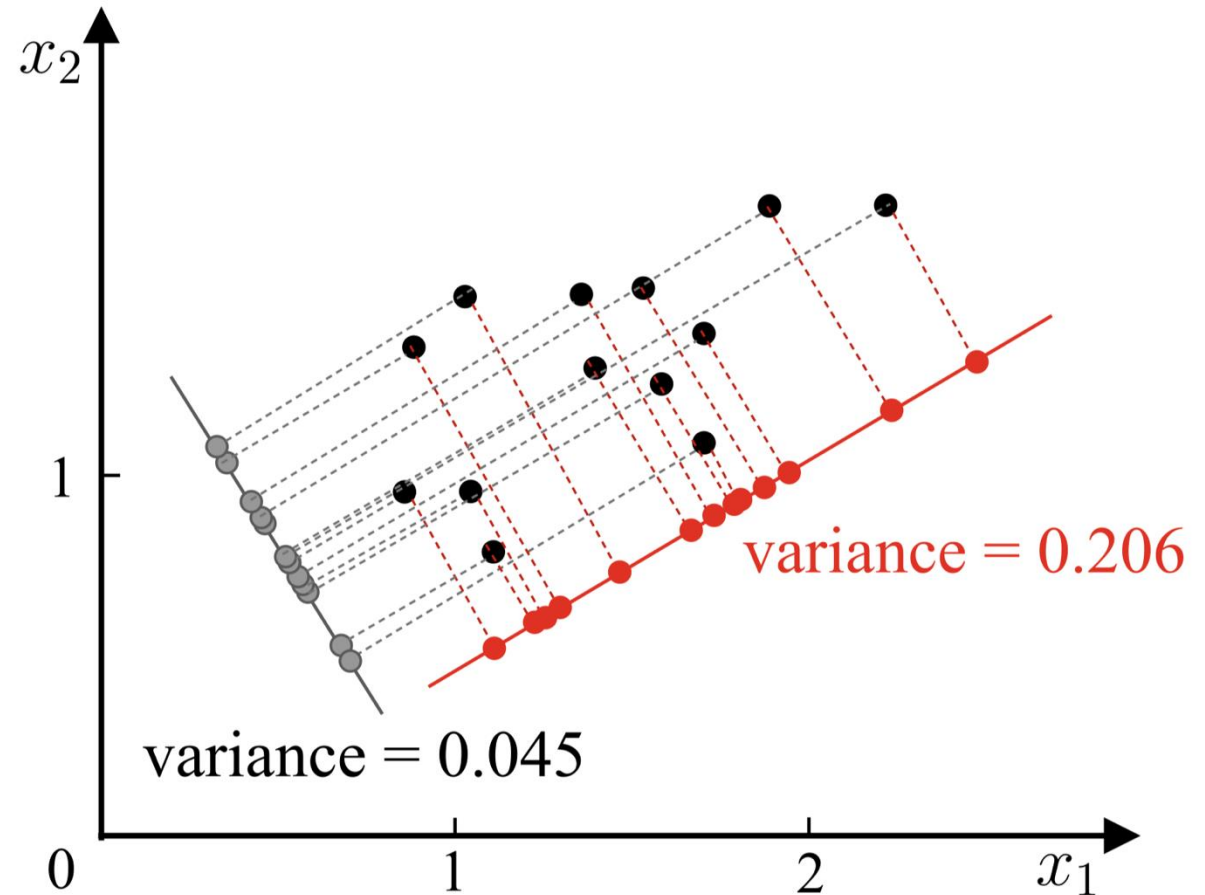
Example: from 2-d to 1-d

- Which project direction should we choose?
 - Red, gray, or ?
- **Max-variance** is the best choice!
 - Data is distributed sparsely
 - Easier for further process



Key idea of Principal Component Analysis (PCA)

- These two are equivalent:
 - **Maximum variance:** the projections of samples onto the hyperplane should stay away from each other.
 - **Minimum reconstruction error:** the samples should have short distances to this hyperplane.



Principal Component Analysis (PCA)

Input: Data set $D = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$;
Dimension d' of the lower dimensional space.

Process:

- 1: Center all samples: $\mathbf{x}_i \leftarrow \mathbf{x}_i - \frac{1}{m} \sum_{i=1}^m \mathbf{x}_i$; 0-mean samples
- 2: Compute the covariance matrix $\mathbf{XX}^T$ of samples;
- 3: Perform eigenvalue decomposition on the covariance matrix $\mathbf{XX}^T$;
- 4: Take the eigenvectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{d'}$ corresponding to the d' largest eigenvalues.

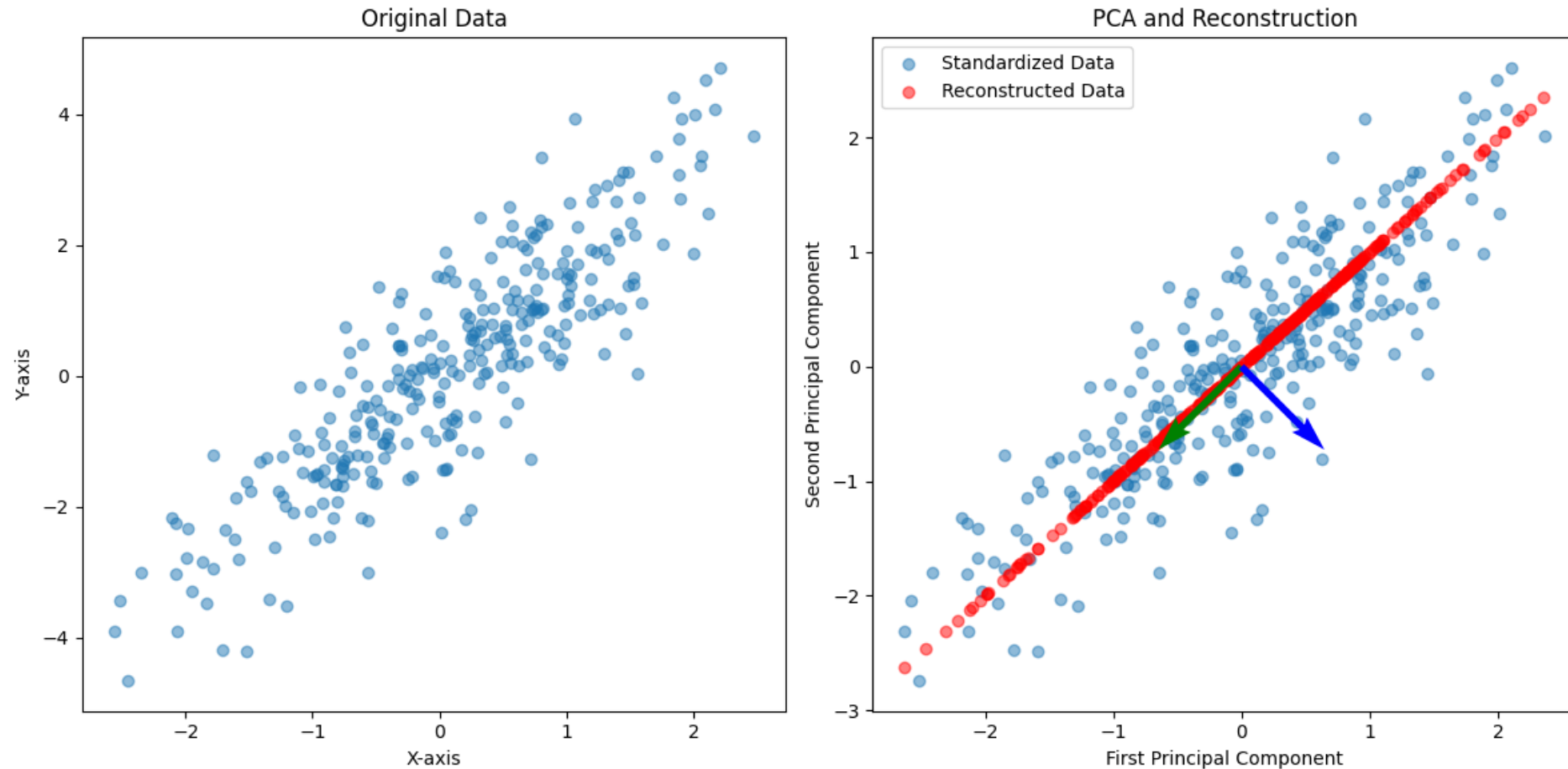
Output: The projection matrix $\mathbf{W}^* = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{d'})$.

Eigenvalue decomposition

$$\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^{-1}$$

The diagram illustrates the eigenvalue decomposition of a matrix $\mathbf{A}$. Matrix $\mathbf{A}$ is represented by a 3x3 grid. It is equal to the product of three matrices: $\mathbf{Q}$, $\mathbf{\Lambda}$, and $\mathbf{Q}^{-1}$. Matrix $\mathbf{Q}$ contains the eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ of $\mathbf{A}$. Matrix $\mathbf{\Lambda}$ contains the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of $\mathbf{A}$. Matrix $\mathbf{Q}^{-1}$ is the inverse of $\mathbf{Q}$. Brackets and labels below the matrices identify them as 'Eigen vectors of A', 'Eigen values of A', and 'Eigen vectors of A' respectively.

Example of PCA on Gaussian data



Different choices of dimension of PCA in image compression



d=1



d=2



d=4



d=8



d=16



d=32



d=64



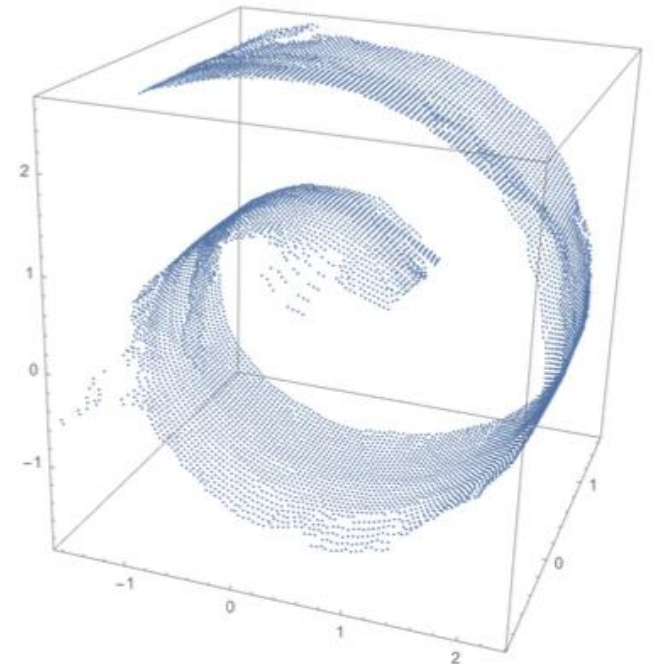
d=100

**Original
Image**

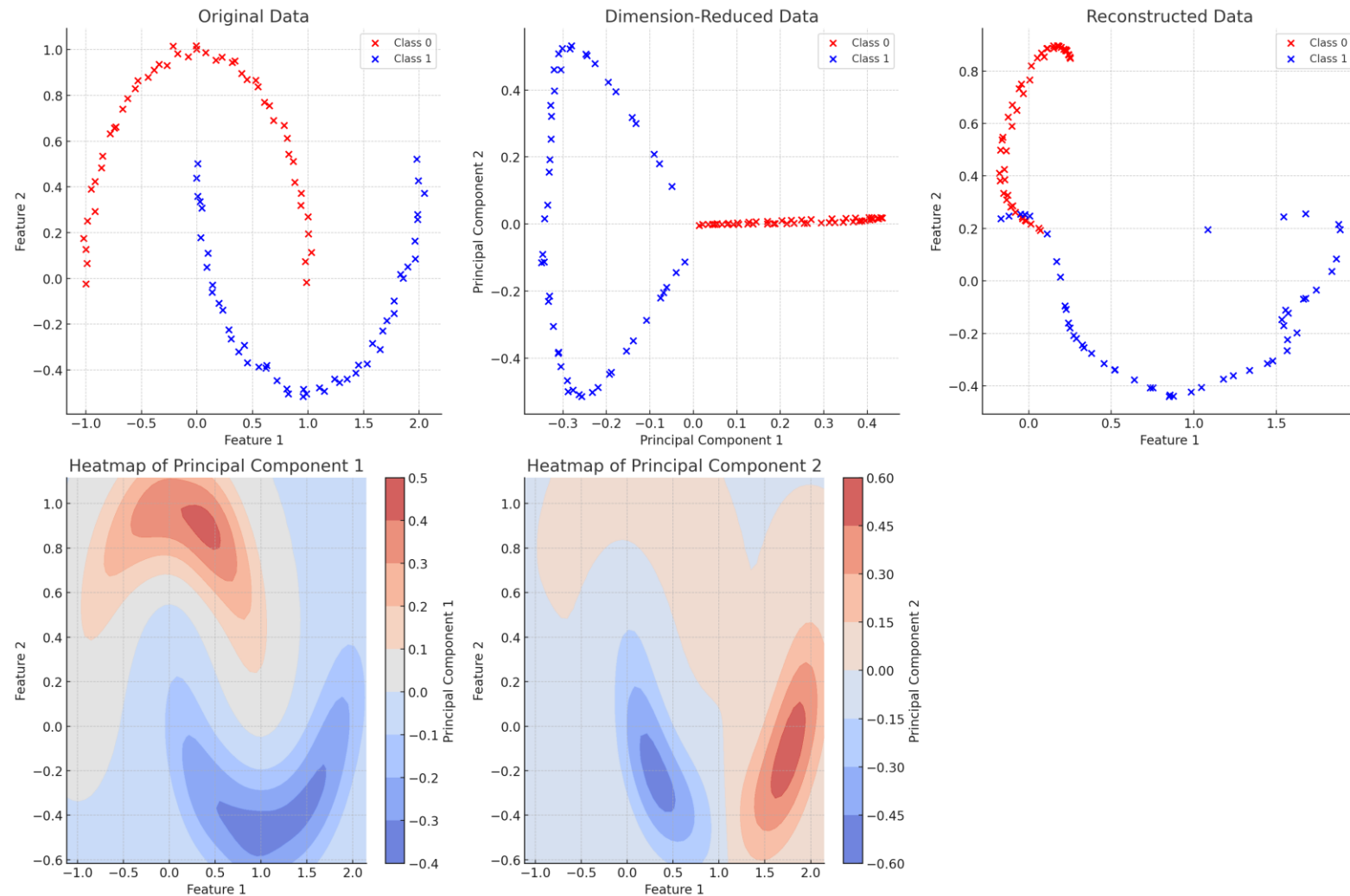


Non-linear dimension reduction

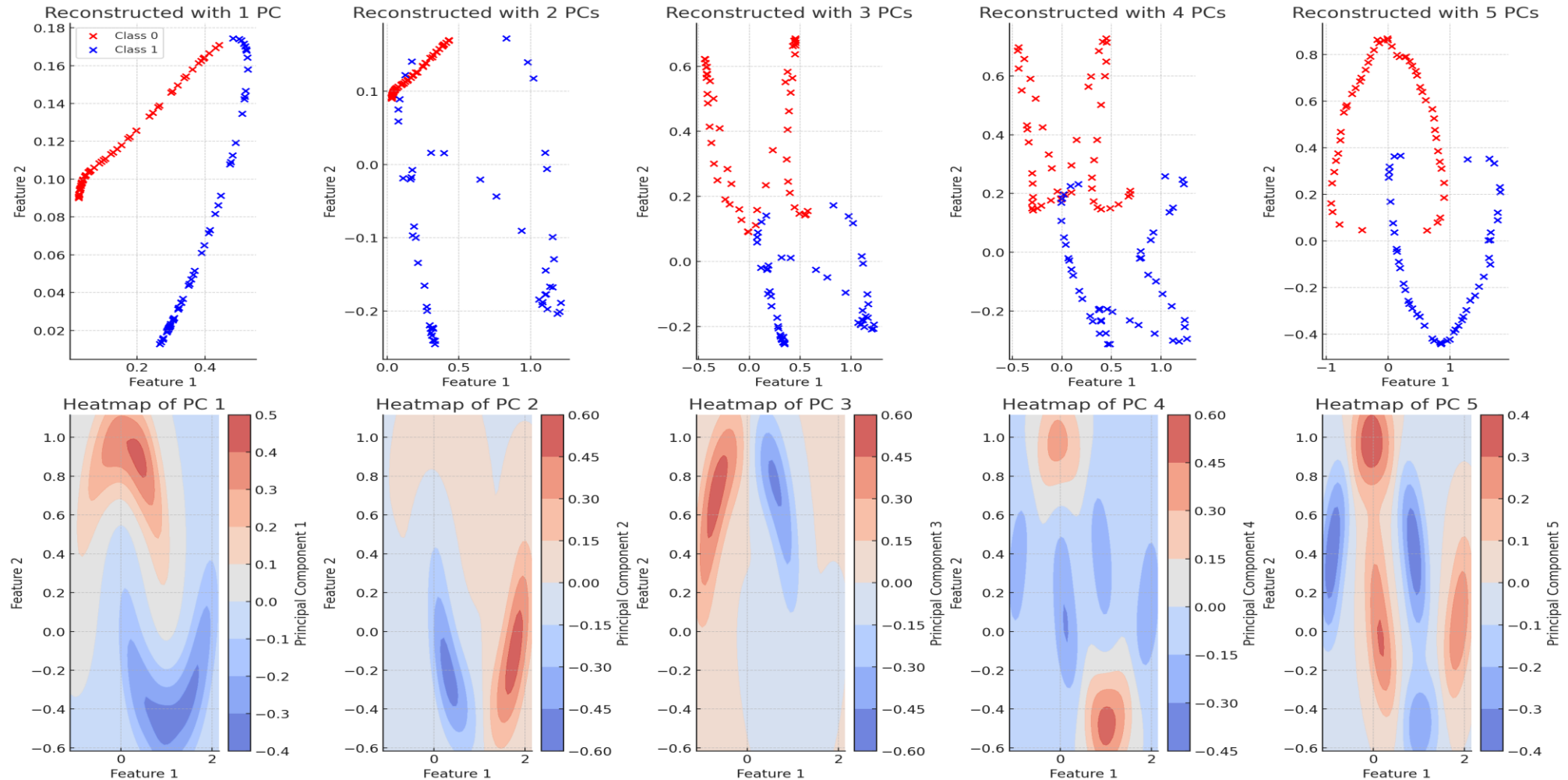
- Mixture of linear subspaces:
 - Subspace clustering
 - Mixture of probabilistic PCAs
 - A combination clustering and dim-reduction
- Kernel PCA
 - Run PCA on the kernel matrix instead of the covariance matrix
- Laplacian Eigenmaps (also the related Isomap)
 - First construct a nearest neighbor graph
 - Then run SVD on the Laplacian matrix of the graph
- Neural approaches:
 - Autoencoders / variational autoencoders
 - Transformers (for data-reconstruction)



Kernel PCA on the two-moon example



Increasing the number of principal components in kernel PCA improves the reconstruction



What's next?

- Wed Apr 15: HW3 due and HW4 out
- Mon Apr 20: Advanced Topic - Decision Making
 - **Not** part of the final exam
 - Most recent hot topics in machine learning!
 - It would be fun!
- Wed Apr 22: Course Review
- Mon Apr 27, Wed Apr 29: Final project presentations
- Mon May 4: HW3 and HW4 Review
- Tue May 12, 3:30-5pm: Final exam